Analytics & Big Data

Session 0: Course logistics

Prof. Dr. Gerit Wagner

(2026-03-23)

Welcome

Prof. Dr. Gerit Wagner

Academic background

- Universität Regensburg: Doctorate
- HEC Montréal: Postdoctoral Fellowship
- Otto-Friedrich-Universität Bamberg: Assistant Professor
- Frankfurt School of Finance & Management: Full Professor

Research interests

- Open, agentic, and boundary-spanning work
- AI-supported knowledge synthesis

Teaching interests

- Analytics and big data
- Programming and software engineering
- IT security
- Digital knowledge-intensive and platform-based work
- Literature review methods

About you

Please answer the three questions. Then choose **one** optional area where you would like to share a bit more.

1. Have you worked in an industry role or completed an internship with an analytics focus?

Optional:

- What was something that felt **very useful** in your analytics work?
- Or where would you have liked to be **better prepared** for a data-related task?

2. Which analytics tools or programming languages have you used?

Optional:

- What was a particularly interesting analytics **challenge** you encountered?
- Or a tool you found especially powerful or **surprising**?

Analytics & Big Data

Analytics & Big Data in the curriculum

1*	Principles of Management	Financial Accounting	Introductionto Programming	Mathematics	Analytical Thinking & Critical Reasoning
2*	Principles of Finance	Microeconomy	Statistics& Probability	Macro-economics	Marketing
3*	Operations Management	Managerial Accounting	Markets, Incentives & Ethical Management	Applied Research Methods	BusinessLaw
4	Corporate Finance	Strategy & Organisation	Concentration		
5	SemesterAbroad &Internship Abroad				
6	Elective	Academic Skills Module	Concentration		
7	Elective	Elective	Elective	Thesis	

What exactly is "analytics"?

Business analytics refers to the computer-supported examination of data using mathematical models to drive decisions and actions within business situations (based on Davenport & Harris 2007).

Why Analytics (and This Course) Matter

💰 Finance — Quantitative Arbitrage at Princeton-Newport Partners

- Early pioneer of applying the Black-Scholes option pricing model in real markets
- Systematic identification of mispriced options through mathematical arbitrage
- Demonstrated how financial models and data could generate consistent trading profits

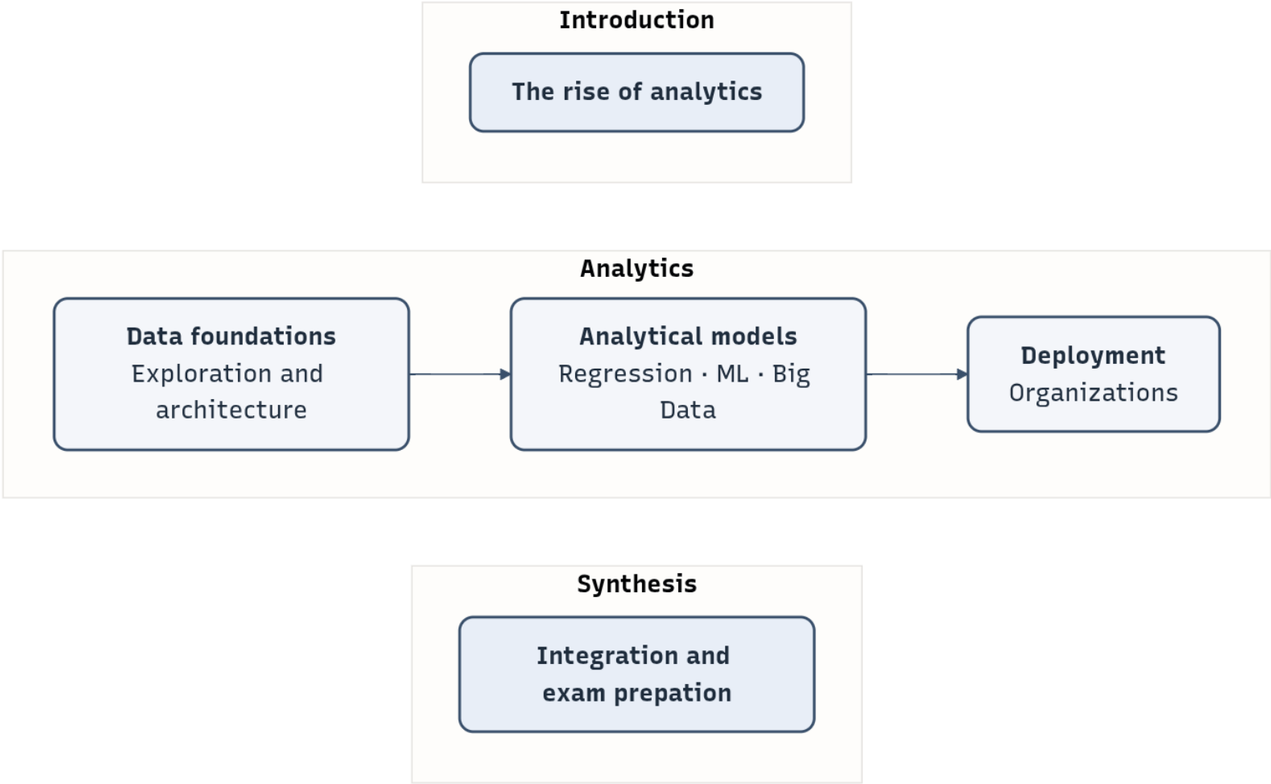
🎬 Entertainment — Personalization at Scale at Netflix

- Recommendation systems shape what millions watch
- Continuous A/B testing and model refinement
- Data directly drives engagement and revenue

⚾ Sports — The Moneyball Revolution at Oakland Athletics

- Advanced metrics replaced traditional scouting intuition
- Data-driven player selection with limited budget
- Competitive advantage through better measurement

Course Architecture



Learning objectives

Knowledge

- **Differentiate** between descriptive, predictive, and prescriptive analytics approaches.
- **Describe** architectures for data warehousing and big data processing, with reference to the underlying data modeling concepts.
- **Explain** fundamental concepts of data analytics, and its role in data-driven organizations.

Skills

- **Select** appropriate analytical methods for transactional or non-transactional data, and for descriptive, predictive, and prescriptive analysis tasks.
- **Apply** analytical methods to real-world datasets, including structuring, transforming, and visualizing data. This also involves training models, evaluating performance, and interpreting analytical results for decision-making.
- **Implement** analytical procedures in Python, using standard data science libraries (`pandas`, `scikit-learn`, `matplotlib`).

Competence

- **Design** analytics solutions aligned with business goals and governance principles.
- **Integrate** analytical technologies into organizational decision-making processes
- **Assess** the operational and strategic implications of data analytics in organizations.

What This Course Does *Not* Cover

- **Engineering of production-grade analytical systems**
  - Software engineering
  - Database design
  - DevOps/MLOps pipelines
  - Scalable application development
- **Statistical theory**
  - Inferential statistics
  - Proofs/derivations
- **Mathematical optimization and simulation models**
  - Scheduling, logistics, queueing
- **Autonomous or agentic decision systems**
  - Robots/agents acting on decisions
  - Reinforcement learning

💡 We will *touch* these areas where needed, but our focus is on **analytical reasoning** and **notebook-based implementations**.

Course logistics

Course logistics

Workload: 150h total (44h in class, remaining: self-study and group project)

Assessment

- Exam: 60 minutes, 60 points
- Group project: written paper (40 pages), 60 points (end of module)

Sessions:

- See overview in Canvas

Contact:

- EMail: g.wagner@fs.de
- Office hours: booking page

Individual circumstances

If you have family responsibilities, religious holidays, health-related matters, or other individual circumstances that may affect your participation or performance, please reach out early. We will work together to find a fair and workable solution.

Materials

Slides and materials

- Presentation slides will be made available for download.
- You are expected to take complementary notes and read the recommended literature.
- Literature and complementary materials will be listed at the end of each lecture.
- Reading of complementary materials depends on your interest and ambition.

Selected literature

- Part 1: Chen et al. (2012), Martínez-Plumed et al. (2019), Sharda et al. (2018)
- Part 2: Raschka (2015), Schutt & O'Neil (2014), Vaisman & Zimányi (2016)
- Part 3: Bengfort et al. (2018). Schmarzo (2016)
- Part 4: Davenport (2006), Vidgen et al. (2017)

You may also be interested in …

Bachelor's theses: See SuSy and additional information on this page.

In SuSy, you can find more information on my primary research topics:

- **Open, Agentic, and Boundary-Spanning Work**
  This area examines how AI agents and open digital infrastructures reshape knowledge work and collaboration across organizational boundaries. It focuses on agentic systems embedded in real work contexts (e.g., Git workflows, handbooks, and repositories) and how they transform coordination, governance, and accountability.
- **AI-Supported Knowledge Synthesis**
  This area investigates how AI supports knowledge synthesis across academic and professional contexts, focusing on transparency, rigor, and traceability in AI-assisted processes. It includes literature reviews and synthesis in digital gardens, second-brain systems, and other knowledge repositories.

References

Bengfort, B., Bilbro, R., & Ojeda, T. (2018). *Applied text analysis with python: Enabling language-aware data products with machine learning*. O'Reilly Media. https://www.oreilly.com/library/view/applied-text-analysis/9781491963036/

Chen, H., Chiang, R. H. L., & Storey, V. C. (2012). Business intelligence and analytics: From big data to big impact. *MIS Quarterly*, *36*(4), 1165–1188. https://www.jstor.org/stable/41703503

Davenport, T. H. (2006). Competing on analytics. *Harvard Business Review*, *84*(1), 98–107. https://cs.brown.edu/courses/cs295-11/competing.pdf

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Raschka, S. (2015). *Python machine learning*. Packt Publishing. https://www.packtpub.com/en-us/product/python-machine-learning-9781783555130

Schmarzo, B. (2016). *Big data MBA*. Wiley. https://doi.org/10.1002/9781119238881

Schutt, R., & O'Neil, C. (2014). *Doing data science*. O'Reilly Media. https://www.oreilly.com/library/view/doing-data-science/9781449363871/

Sharda, R., Delen, D., & Turban, E. (2018). *Business intelligence, analytics, and data science: A managerial perspective*. Pearson. https://elibrary.pearson.de/book/99.150005/9781292727530

Vaisman, A., & Zimányi, E. (2016). *Data warehouse systems: Design and implementation*. Springer. https://doi.org/10.1007/978-3-662-65167-4

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